MILITARY SPECIFICATION
CASE, CARTRIDGE, 57MM, M30A1B3

1. SCOPE

1.1 This specification covers Case, Cartridge, 57mm, M30A1B3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For

STANDARDS

Federal

Federal Test Method Standard No. 141	- Paint, Varnish, Lacquer and Related Materials, Method of Inspection, Sampling and Testing
Federal Test Method Standard No. 151	- Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

F11738308	- Case, Cartridge, 57mm, M30A1B3
F7548235	- Cartons, Ammunition for Interplant Shipment of Steel Cartridge Cases, 37mm to 120mm, inclusive
IEL 11738308	- Inspection Equipment List for Case, Cartridge 57mm, M30A1B3

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th St., New York, N.Y.)

3. REQUIREMENTS

3.1 General.- The cartridge case shall comply with all the requirements of the applicable drawings, referenced specifications and the following:

3.2 Material.- The supplier shall be responsible for selecting from the classification list, contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.3 Mechanical Properties.- The mechanical properties specified on the applicable drawing and referenced specification shall be uniform throughout each section.

3.3.1 Thermal Treatment.- The cartridge case body shall be subjected to a thermal treatment after the assembly operation by heating above the upper critical temperature point, quenching and tempering to the required properties.

3.4 Stress Relief.- The body of the cartridge case shall be adequately stress relieved at the times and temperature suitable for development of required properties.

3.5 Strength of Brazed Joint.- The head and the tube of the cartridge case shall be joined in the manner specified on the applicable drawing. The joint shall withstand a pull of 3000 pounds, minimum, without rupture.

3.6 Strength of Weld Zone.- The weld zone of the cartridge case body after final assembly and heat treatment shall be as strong as any other parallel section of the body.

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3.7 Quality of Weld Zone.- The weld zone joint shall be complete and contain no cracks.

3.8 Protective Coating.- Protective coating shall be as specified on the applicable drawing and referenced specifications and shall apply to all surfaces of the cartridge case regardless of contour configuration.

3.9 Maximum Profile.- Maximum profile of the cartridge case shall comply with the requirements of the applicable drawing.

3.10 Functional.- The cartridge case shall be subjected to a functional test in the following phases:

Excess Pressure Phase
Cold Temperature Phase
Excess, Excess Pressure Phase (for information only)
Recoil Momentum Phase (for information only)

3.10.1 Soundness.- After firing, the cartridge case shall be free of cracks, splits, bulges or ruptures.

3.10.2 Extraction.- After firing, the cartridge case shall be extracted by the service extractors when the weapon is operated in the normal manner (See 6.3).

3.10.3 Primer Retention.- The cartridge case shall retain its primer during firing and extraction.

3.11 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Metal Defects.- The cartridge case shall be free from cracks, die scratches, flutes, folds, dents, laminations and porosity.

3.11.2 Surface Finish.- The requirements for surface finish shall be as specified on the applicable drawing.

3.11.3 Burr.- No cartridge case shall have a burr which might interfere with the assembly of the primer or projectile or which might be injurious to personnel handling the item.

3.11.4 Foreign Matter.- No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First Article Sample.

4.2.1 Initial Production Sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 229 cartridge cases as follows:

209 Cartridge Cases - Coated
20 Cartridge Cases - Uncoated

205 of the above coated cartridge cases shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and Test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial Production Sample Failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection Provisions.

4.3.1 Lot.

4.3.1.1 Submission of Product.- The product shall be submitted in accordance with MIL-STD-105.

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4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. All nonconforming material shall be rejected. Four power (4x) magnification may be used in performing visual examination for such defects as cracks, burr and foreign matter. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Case, Cartridge, Drawing F11738308 (prior to coating).-

	Defect	Method of Inspection
CRITICAL		
	None Defined	
MAJOR		
101.	Primer hole counterbore not functionally concentric with primer hole	Gage
102.	Primer hole not functionally concentric with body	Gage
103.	Outside diameter of mouth not functionally concentric with basic diameter and flange	Gage
104.	Flange not functionally concentric with primer hole	Gage
105.	Diameter of mouth, inside	Gage
106.	Diameter of flange	Gage
107.	Diameter of primer hole	Gage
108.	Taper in primer hole, incorrect	Gage
109.	Thickness of flange	Gage
110.	Thickness of wall at mouth for min. length	Gage
111.	Thickness of wall of protuberances, min.	Gage
112.	Thickness of sidewall of head	Gage
113.	Wall thickness of tube rear of protuberances	Gage
114.	Height of Protuberances	Gage
115.	Depth of primer hole counterbore, min.	Gage
116.	Length from bottom of counterbore to basic diameter, min.	Gage
117.	Length from rear face to basic diameter, max.	Gage
118.	Length from rear face to front of head	Gage
119.	Weld not flush with inside diameter of mouth for indicated length	Gage
120.	Angle of Protuberances	Gage

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121.	Diameters inside (min.) forward and rear of protuberances	Gage
122.	Metal defective	Visual

MINOR

201.	Diameter of primer hole counterbore	Gage
202.	Diameter of undercut forward of flange for max. length (if applicable)	Gage
203.	Length, total	Gage
204.	Thickness through head	Gage
205.	Location of first row circumferential holes	Gage
206.	Spacing and alignment of holes, improper	Visual
207.	Row of hole(s) missing	Visual
208.	Welding extended into perforated area	Visual
209.	Surface finish, improper	Visual (see 6.2)
210.	Radii or chamfers, missing or incorrect	Visual (see 6.2)
211.	Marking missing, incorrect or illegible	Visual
212.	Burr	Visual
213.	Foreign matter	Visual

4.3.2.2 Case, Cartridge, Drawing, F11738308 (after coating).-

Defect	Method of Inspection

CRITICAL

None Defined

MAJOR

101.	Dents	Visual
102.	Metal defective	Visual

MINOR

201.	Protective coating inadequate or damaged	Visual
202.	Varnish in primer hole	Visual
203.	Compound missing from primer hole	Visual
204.	Foreign matter	Visual

4.3.2.3 Carton, Packed (prior to sealing) Drawing F7548235.-

Defect	Method of Inspection

CRITICAL

None Defined

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MAJOR

None Defined

MINOR

201.	Support missing or damaged	Visual
202.	Separator missing or damaged	Visual
203.	Improper assembly	Visual
204.	Incorrect number in carton	Visual

4.3.2.4 Carton, Packed (after sealing) Drawing F7548235.-

Defect	Method of Inspection

CRITICAL

None Defined

MAJOR

None Defined

MINOR

201.	Carton damaged, cut or gouged through all layers	Visual
202.	Carton not completely sealed	Visual
203.	Contents loose	Manual
204.	Marking missing, incorrect or illegible	Visual

4.3.3 Inspection Testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Maximum Profile Gage (a major defect).- Each cartridge case shall be gaged in a government certified maximum profile gage. Failure of the cartridge case to comply with the requirements of 3.9 shall be cause for rejection of the cass.

4.3.3.2 Length, min. of inside diameter at mouth (a major defect).- Each cartridge case shall be gaged at the mouth for compliance with the requirements of the applicable drawing, failure of the cartridge case to comply with the requirements shall be cause for rejection of the cartridge case.

4.3.3.3 Mechanical Properties.

4.3.3.3.1 Hardness.- Ten cartridge cases from the initial production sample and 4 cartridge cases from each production lot shall be subjected to this test. Failure of one or more cartridge cases to comply with the requirements of the applicable drawing shall be cause for rejection of the sample or lot.

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4.3.3.3.2 Thermal Treatment (a major defect).- Each cartridge case shall be subjected to a hardness comparator test. Failure of the cartridge case to comply with the requirements of 3.3.1 or indications that the cartridge case failed to harden shall be cause for rejection of the cartridge case.

4.3.3.4 Strength of Brazed Joint.- (Head to tube) (A major defect) Ten cartridge cases from the initial production sample and 4 cartridge cases from each production lot shall be subjected to this test. Failure of one or more cartridge cases to comply with 3.5 shall be cause for rejection of the sample or lot.

4.3.3.5 Soundness of Weld.- (A critical defect) - Each cartridge case, after final expansion but prior to application of varnish, shall have both surfaces of the weld examined visually to determine compliance with 3.6 and 3.7. Failure to comply with the requirements shall be cause for rejection of the cartridge case.

4.3.3.6 Protective Coating.- The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.3.3.7 Functional.- The following proving ground tests will be performed by the government:

4.3.3.7.1 Initial Production Sample.- The quantities listed in Table I will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table III.

TABLE I

Test Phase	Quantity
Excess Pressure Phase	100
Cold Temperature Phase	50
Excess, Excess Pressure Phase (information only)	50
Recoil Momentum (information only)	5

TABLE II

Defect Characteristic	Sample - Accept
1. Splits or cracks	0
2. Extraction	0
3. Primer retention	0

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4.3.3.7.2 Production Lot.- The quantities listed in Table III will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table IV. The lot will be considered eligible for retest, using double the number of samples, provided that after firing, the rejection criteria has not been attained.

TABLE III

Test Phase	Quantity
Excess Pressure Phase	40
Recoil Momentum (information only)	5

TABLE IV

Defect Characteristic	First Sample		Cumulative Sample	
	Accept	Reject	Accept	Reject
1. Splits or cracks	0	1	Not permitted	
2. Extraction	0	1	Not permitted	
3. Primer retention	0	2	1	2

4.3.4 Inspection Equipment.- Index of inspection equipment list, IEL 11738308 identified the inspection equipment required to perform the examinations and tests prescribed herein.

4.4 Test Method and Procedures.

4.4.1 Hardness.- This test shall be performed on the locations specified by the applicable drawing in accordance with the procedures prescribed for Rockwell Hardness testing by Federal Test Method Standard No. 151.

4.4.1.1 Thermal Treatment.- This test shall be performed on a hardness comparator of a nondestructive type approved by the Contracting Officer.

4.4.2 Strength of Brazed Joint (Head to tube)- The cartridge cases selected in accordance with 4.3.3.4 shall be mounted on an approved test fixture. The pressure shall be applied gradually until the required load has been attained. Upon removal from the test fixture, the joint will be visually examined for crack(s) in the brazed area or deformation of the cartridge case in the heat affected zone.

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4.4.3 Soundness of Weld.- Prior to coating, the weld seam on the outside surface shall be visually examined for crack(s) or other discontinuities. The inside surface of the weld seam shall be visually examined for crack(s) or other discontinuities, utilizing a 4-power borescope.

4.4.4 Protective Coating.- The methods and procedures for performing these tests shall be in accordance with the requirements of the applicable specification.

4.4.5 Maximum Profile Gage.- The cartridge case, after application of protective coating shall freely enter the government certified maximum profile gage. The gate of the gage shall close without interference with the cartridge case head.

4.4.6 Mouth, Inside Diameter.- After maximim profile gaging, the mouth of the cartridge case shall be gaged for the minimum length required by the applicable drawing. The gage shall enter and be withdrawn freely without damage to the varnish.

4.4.7 Functional.

4.4.7.1 Initial Production Sample.

4.4.7.1.1 Excess Pressure Phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees Fahrenheit (F) to plus 75 degrees F, and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.7.1.2 Cold Temperature Phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce a chamber pressure of 6700 pounds per square inch (psi) to 7700 psi piezo equivalent pressure, when temperature conditioned at plus 70 degrees F. The complete round will be temperature conditioned at minus 35 degrees F., to minus 45 degrees F., and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.7.1.3 Excess, Excess Pressure Phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the maximum chamber pressure permitted by the proving ground. The complete round will be temperature conditioned at plus 65 degrees F., to plus 75 degrees F., and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

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4.4.7.1.4 Recoil Momentum Phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the service velocity and pressure for the complete round. The recoil momentum will be recorded.

4.4.7.2 Production Lot.

4.4.7.2.1 Excess Pressure Phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees F., to plus 75 degrees F., and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.7.2.2 Recoil Momentum Phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the service velocity and pressure for the complete round. The recoil momentum will be recorded.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A - Not applicable

5.1.2 Level B - Not applicable

5.1.3 Level C - The cartridge case shall be packed in accordance with the requirements of Drawing F7548235.

5.2 Packaging and Packing.- Packaging, packing and marking shall be in accordance with MIL-STD-1169.

5.3 Data Card.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Prior to fabrication of an initial production sample, the contractor upon approval of the Contracting Officer, may make and submit for evaluation up to 5 process control lots. These process control lots shall consist of not less than 10 nor more than 25 cartridge cases each. The contractor shall furnish all information pertaining to the manufacture of the lots to the Contracting Officer. All costs for testing these process control lots will be borne by the government.

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6.2 Visual Examination Qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Extraction.- The term "Extraction" shall mean that the fired cartridge case is loosened and pulled to the rear of the weapon sufficiently to permit removal by hand.

Custodian

Army - MU

Preparing activity

Army - MU (FA)

Project No. 1310-A201

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. *Fold on lines on reverse side, staple in corner, and send to preparing activity.* Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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